

DOCUMENT 05

Failure Engineering & Mission Assurance

Complete FMEA · Fault Trees · Reliability · Emergency Procedures

Aerospace-grade failure engineering covering complete FMEA for all seven drone subsystems, fault tree analysis for critical top events, reliability block diagrams with quantified failure rate budgets, mission assurance targets, and detailed emergency procedures for every identified failure class including GNSS spoofing, mesh partition, and docking failure.

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1. FMEA METHODOLOGY

Failure Modes and Effects Analysis (FMEA) is conducted per MIL-STD-1629A methodology. Risk Priority Number (RPN) = Severity (S) × Occurrence (O) × Detectability (D), each scored 1–10. Critical items (RPN > 100 or Severity ≥ 9) require documented mitigation before flight. The complete FMEA covers seven subsystem groups: propulsion, power/energy, navigation/control, communication/coordination, docking, structural/mechanical, and AI/compute.

S/O/D Level	Score	Severity Definition	Occurrence Definition	Detectability Definition
Catastrophic/Frequent/Undetected	9–10	Loss of vehicle or injury risk	Likely per 100 flights	No warning before failure
Critical/Occasional/Poor	7–8	Mission abort; vehicle damage	Likely per 500 flights	Detected only after failure occurs
Major/Remote/Moderate	5–6	Degraded mission; safe return	Possible per 1,000 flights	Usually detected by monitoring
Minor/Unlikely/Good	3–4	Reduced performance; continues	Rare per 10,000 flights	Detected by standard checks
Negligible/Improbable/Excellent	1–2	No mission impact	Extremely rare	Always detected immediately

Table 1. FMEA scoring criteria.

2. PROPULSION SUBSYSTEM FMEA

ID	Failure Mode	Root Cause	Effect on Mission	S	O	D	RPN	Mitigation
P01	Motor phase open-circuit	ESC FET fail; winding break	Loss of one motor; drone yaws; emergency descent	8	3	3	72	3-motor emergency land mode in FW; pre-flight motor run-up current check
P02	Two co-diagonal motors fail	Shared arm short; lightning	Uncontrollable roll; crash	10	1	1	10	Per-arm power fusing; physical arm separation
P03	ESC thermal shutdown >90°C	Sustained >80% throttle; cooling fail	Motor cut; hard landing	7	3	5	105	CRITICAL — thermal monitoring; auto-throttle-back at 80°C; mandatory fan at 35°C+
P04	Prop strike/fracture	Obstacle; ground contact	Vibration spike; motor damage	8	3	6	144	CRITICAL — carbon props; visual inspection each flight; prop guards for cluttered environments
P05	Motor bearing seizure	Contamination; wear (>200 hr)	Motor halt; vibration spike	8	2	4	64	200 hr replacement schedule; pre-flight vibration check via IMU
P06	ESC firmware lock-up	Bug; EMI spike	Motor cut on one ESC	6	2	5	60	BEL_32/AM32 watchdog; hardware reset pin; dual redundant ESC option Phase 2
P07	Prop nut loosening	Insufficient torque on install	Prop loss in flight; crash	9	2	7	126	CRITICAL — 1.2 N·m torque spec; threadlock on self-lock nuts; check post-10-flight
P08	Motor winding insulation fail	Moisture ingress; age	Inter-winding short; motor burn	7	1	4	28	Conformal coat windings; inspect annually; smoke detector in motor bay

Table 2. Propulsion FMEA. Items with RPN ≥ 100 are highlighted Critical.

3. POWER AND ENERGY SUBSYSTEM FMEA

ID	Failure Mode	Root Cause	Effect	S	O	D	RPN	Mitigation
E0 1	Battery thermal runaway	Overcharge; physical damage; deep discharge	Fire; structural damage; crash	10	1	2	20	LiFePO4 preferred; BMS cutoff 60°C; crash sensor disconnects pack
E0 2	FC brownout under AI load	Voltage sag from unified power rail	FC reset; attitude hold lost	8	1	2	16	SOLVED: Isolated PM02D rail. 0% brownout in all post-fix trials.
E0 3	Battery connector arcing	Contaminated XT60; high-impedance contact	Connector damage; potential fire	7	3	5	105	CRITICAL — antispark XT60; inspect pre-flight; gold-plated dock pogo pins
E0 4	DC-DC converter failure (PM02D)	Overvoltage; component ageing	FC power lost; uncontrolled descent	9	1	3	27	Capacitor hold-up 200 ms; redundant 5V rail Phase 2; annual replacement
E0 5	Battery deep discharge below 3.0V/cell	RTH not triggered; mission timer fail	Cell damage; short-circuit risk	6	2	5	60	Hardware BMS cutoff 3.0V/cell independent of FC; redundant SOC alert
E0 6	Main power rail short	Wire chafe on frame; loose terminal	Complete power loss; crash	10	1	2	20	100A blade fuse on main bus; wire loom management; strain reliefs
E0 7	Pogo pin pitting/corrosion	Repeated arcing; moisture	High contact resistance; slow/no charge	5	4	6	120	CRITICAL — SSR closes only on R-signature; gold plating; 300-cycle inspection
E0 8	Battery cell imbalance >200 mV	Unbalanced cells; age	Reduced usable capacity; early cutoff	4	4	7	112	Active BMS balance circuit; balance after every cycle; replace at >200 mV

Table 3. Power subsystem FMEA.

4. NAVIGATION AND CONTROL FMEA

ID	Failure Mode	Root Cause	Effect	S	O	D	RPN	Mitigation
N0 1	GPS signal loss — jamming	Adversarial; industrial RF; multipath	Mission hold; cooperative loc fallback	7	3	7	147	Dual-band GPS option; velocity consistency check; FEDERATED_EKF state
N0 2	GPS spoofing attack	Deliberate false signal	Navigate to wrong position; potential LOA	9	1	3	27	Velocity vs IMU consistency check (>3 m/s discrepancy = spoof flag); RTH on flag
N0 3	IMU accelerometer bias drift	Temperature change; vibration	Attitude error; SOC estimation degrades	6	3	5	90	In-flight temp compensation; 200 hr IMU replacement; dual-IMU cross-check
N0 4	Compass interference	Power wiring; motors; metal objects	Heading error; position circle drift	6	4	5	120	Mast-mounted compass; compass health check in pre-flight; dual-source heading
N0 5	VIO feature tracking failure	Low texture; low light; motion blur	GPS-denied nav degrades	6	4	5	120	Minimum 200 lux lighting for VIO ops; texture-rich environment; fallback to inertial hold
N0 6	Cooperative loc error >2 m	C1 heading drift; large separation	Worker position jumps; instability	8	2	3	48	Dual-antenna RTK on C1 (<0.1° heading); UWB cross-check; covariance gating

ID	Failure Mode	Root Cause	Effect	S	O	D	RPN	Mitigation
N07	PID tuning instability	Wrong gains after mass change	Oscillations; reduced stability margin	6	3	6	108	Mass-dependent gain schedule; Monte Carlo gain robustness verification
N08	Barometer icing (cold ops)	Condensation in port; ice crystals	Altitude hold failure; uncontrolled climb	7	2	3	42	Heated baro port for sub-5°C ops; rate-of-climb anomaly detection

Table 4. Navigation and control FMEA.

5. COMMUNICATION AND COORDINATION FMEA

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Mitigation
C01	LoRa link loss (primary)	Obstacle; range exceeded; hardware fault	GCS loses telemetry; operator cannot issue commands	7	3	8	168	2.4 GHz DSSS backup; drone continues stored mission autonomously for 10 min; RTH after
C02	Both links lost simultaneously	Wideband RF jamming; simultaneous faults	Full communication blackout	8	2	2	32	Stored waypoint mission for 10 min; RTH to home position; log all telemetry to SD
C03	ESP-NOW mesh partition	Physical separation; obstacle; congestion	Fleet coordination fragments; workers lose relay	6	3	5	90	Relay node drones; sub-swarm autonomous CBBA-SOC; RTH if isolated >60 s
C04	TDMA slot collision	Timing drift; node addition	Garbled packets; intermittent latency spikes	4	4	5	80	CSMA fallback; slot re-sync on collision detection; RTH priority always reserved
C05	Command node failure mid-mission	Battery depleted; hardware fault	Workers lose cooperative loc; lose task assignment	8	2	4	64	Federated EKF activates; CBBA-SOC runs autonomously on each worker; RTH at SOC<30%
C06	ESP-NOW node limit exceeded	Fleet scales >20 nodes	Mesh coordination fails for excess nodes	6	2	4	48	Hard limit: max 15 active nodes per ESP-NOW network; migrate >15 to Wi-Fi mesh
C07	GCS software crash	OS error; hardware fault on GCS PC	Ground monitoring lost; operator loses situation awareness	5	3	6	90	Watchdog on GCS; drones continue autonomous mission; SMS/push alert to operator phone

Table 5. Communication subsystem FMEA.

6. DOCKING SUBSYSTEM FMEA

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Mitigation
D01	Pogo pin no-contact	Debris; spring fatigue; misalign	No charge; worker stranded at dock with draining battery	6	4	7	168	3 auto-retry attempts; pin cleaning procedure; 500-cycle spring replacement
D02	IR beacon lamp failure	LED burn-out; moisture in enclosure	Approach guidance lost; docking suspended	7	2	5	70	4 redundant LEDs; 3-of-4 fallback; asymmetric signal triggers beacon fault alert

ID	Failure Mode	Cause	Effect	S	O	D	RPN	Mitigation
D03	Drone overshoot in descent	Wind gust; IR gain too high	Prop contact with dock structure; potential crash	9	2	5	90	Speed ramp: ≤ 0.3 m/s vertical below 2 m; current limiter on descent rate
D04	Magnetic keel not captured	Misalignment >50 mm; funnel debris	Worker hovers, battery drains; mission abort	7	3	5	105	Funnel corrects ± 100 mm; visual camera fallback; clear funnel during maintenance
D05	SSR fails open	Component fail; firmware fault	No charging; worker stranded	6	2	4	48	Physical fuse as backup path; manual override command from GCS
D06	High-wind approach failure	Wind >12 m/s	Worker cannot stabilise for approach	7	3	4	84	Mission abort threshold 12 m/s; anemometer at dock; no-dock weather flag auto-set
D07	Contact arcing at dock	High voltage differential; contamination	Pogo pin pitting; potential ignition	9	1	1	9	SSR only closes on R-signature match; pre-contact voltage equalisation circuit

Table 6. Docking subsystem FMEA.

7. FAULT TREE ANALYSIS

7.1 Top Event: Loss of Aircraft (LOA)

Fault tree analysis for Loss of Aircraft identifies minimal cut sets — the smallest combinations of basic failure events that individually cause the top event. Three cut sets with highest risk:

Cut Set	Basic Events	Combined RPN	P(per flight)	Mitigation Effect
CS1 — Propulsion catastrophe	Prop fracture (P04) AND emergency land mode fail	$144 \times 15 =$ complex	$\sim 2 \times 10^{-4}$	Carbon prop + pre-flight check reduces to $\sim 5 \times 10^{-5}$
CS2 — Navigation-comms dual fail	GPS loss (N01) AND both comm links fail (C02)	$147 \times 32 =$ complex	$\sim 1 \times 10^{-4}$	Stored mission reduces probability to $\sim 2 \times 10^{-5}$
CS3 — Thermal shutdown + low altitude	ESC thermal (P03) AND altitude < 10 m AGL	$105 \times$ situation	Operational constraint	Min 20 m AGL during missions; thermal pre-monitoring
CS4 — Battery thermal runaway	Cell overhear (E01) AND BMS fail	$10 \times 25 =$ complex	$\sim 5 \times 10^{-4}$	LiFePO4 cells + crash disconnect to $\sim 5 \times 10^{-5}$

Table 7. Fault tree minimal cut sets for Loss of Aircraft.

7.2 Mission Failure Top Events (No LOA)

Battery depletion before dock: Probability = $P(\text{SOC model error } >15\%) \times P(\text{RTH not triggered})$. With EKF SOC error 0.9% RMS (CFRP) and RTH at 40% SOC: model error of 15% is $>16\sigma$ event. $P(\text{mission failure from battery depletion}) < 0.001\%$ per flight.

Docking failure × 3 attempts: At 95% success per attempt: $P(3 \text{ consecutive fails}) = 0.05^3 = 1.25 \times 10^{-4}$ per rotation event in calm conditions. At 8 m/s wind (85% per attempt): $P(3 \text{ fails}) = 0.15^3 = 3.4 \times 10^{-3}$ per event. With 14 rotation events per 7-hour mission: $P(\text{at least one 3-fail sequence}) \approx 4.7\%$ at 8 m/s. This is above the acceptable 1% threshold — outdoor wind docking must target $\geq 92\%$ per attempt.

8. RELIABILITY ESTIMATES AND TARGETS

Subsystem	MTBF Target	Current Estimate	Basis	Action If Not Met
Motor (T-MOTOR F90)	500 hr	500 hr (manufacturer)	Manufacturer data	200 hr bearing replacement schedule
FOC ESC (F55A Pro II)	400 hr	400 hr (estimated)	Thermal derating analysis	Active cooling mandatory at >35°C ambient
Flight Controller (Pixhawk 6C)	2,000 hr	2,000+ hr (STM32H7 space-grade heritage)	Holybro data	Annual replacement recommended
Li-ion 18650 battery	200 cycles (capacity)	500+ cycles (cell spec)	Cell manufacturer data	Replace at <80% capacity; cycle log mandatory
Pogo-pin contacts	1,000 mating cycles	1,000+ (gold plating)	Contact resistance data	Replace springs at 500 cycles; pins at 1,000
IR beacon LEDs	25,000 hr	50,000 hr (OSRAM SFH4557)	Manufacturer data	Replace with beacon housing at 25,000 hr (2.8 years continuous)
Complete worker drone	>200 hr	Not yet measured	Weakest component = motor	Establish fleet-level reliability database after Phase 1

Table 8. Subsystem reliability targets and estimates.

9. EMERGENCY PROCEDURES — COMPLETE SET

9.1 Motor Failure In Flight

- FC detects motor dropout via RPM monitoring or current anomaly within 20 ms
- FC activates 3-motor emergency mode: increased collective and differential torques compensate
- Command node receives MOTOR_FAIL alert; removes worker from active fleet roster
- Worker executes controlled descent to nearest pre-programmed safe landing zone
- Operator notified via GCS alert + SMS; dispatches recovery team
- CBBA-SOC redistributes failed drone coverage area; mission continues with reduced fleet

9.2 Complete Communication Blackout

- Both LoRa and DSSS links absent >10 s: drone enters autonomous stored-mission mode
- Completes current waypoint segment, then executes return-to-home at cruise altitude
- RTH altitude: max(current altitude, 25 m AGL) to clear obstacles
- At home position: land; disarm motors; flash navigation lights for visibility
- All actions logged to onboard SD; downloaded via USB at recovery
- Operator: do not re-enter airspace until drone has landed; await recovery signal on GCS

9.3 GNSS Spoofing Response

- IMU integrates acceleration independently; compare GPS velocity vs IMU-integrated velocity
- Threshold: if $|\text{GPS_vel} - \text{IMU_vel}| > 3 \text{ m/s}$ for >2 s: GPS_SPOOF flag raised
- FC ignores GPS input; switches to COOPERATIVE localisation state (command node)
- If command node also shows inconsistent position data: enter INERTIAL_HOLD
- Mission paused; drone holds position; operator alerted; manual RTH issued via LoRa

9.4 Docking Failure — Three Consecutive Attempts

- Attempt 1 fails (no pogo contact within 5 s of magnetic capture): ascend 3 m, re-approach
- Attempt 2 fails: ascend 5 m; broadcast DOCK_FAIL alert; command node alerts GCS operator

- Attempt 3 fails: execute LOITER_TO_LAND — land at pre-programmed safe flat area within 50 m of dock
- Operator dispatches manual battery swap or vehicle recovery
- Post-event: inspect pogo pins, keel rail, funnel geometry; clean all contact surfaces
- Root cause log entry mandatory before next docking attempt

9.5 Battery Critical Low Mid-Mission (SOC < 15%)

- SOC < 15%: hardware BMS initiates LAND_NOW regardless of mission state
- FC immediately executes emergency descent at current position (not RTH — insufficient energy)
- Broadcast EMERGENCY_LAND {position, SOC} on all radio channels
- All other fleet drones hold position or increase altitude to clear landing zone below
- After landing: motors disarm automatically; flashing lights activate for 30 min
- This scenario indicates RTH trigger at 40% SOC failed — mandatory investigation